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TRIAL 2: WEAPON SHAFT BEND

● PASS VERIFIED

Does a 10 mm 4140 prehard dead shaft take a permanent bend over 0.1 mm on a square 600 J hit?

SHAFT BENDS > 0.1 MM? FAIL

YES

permanent bend 0.139 mm, peak shaft load 19.6 kN

SHAFT BEND [GOVERNING] FAIL

SF 0.722

P(fail) 67% | SF P5 0.0531

Margin to each user-defined failure mode



Trial 2 weapon shaft bend: VERIFIED | PASS 11 WARN 1 FAIL 0 SKIP 0 | 167 runs, 155.2 s

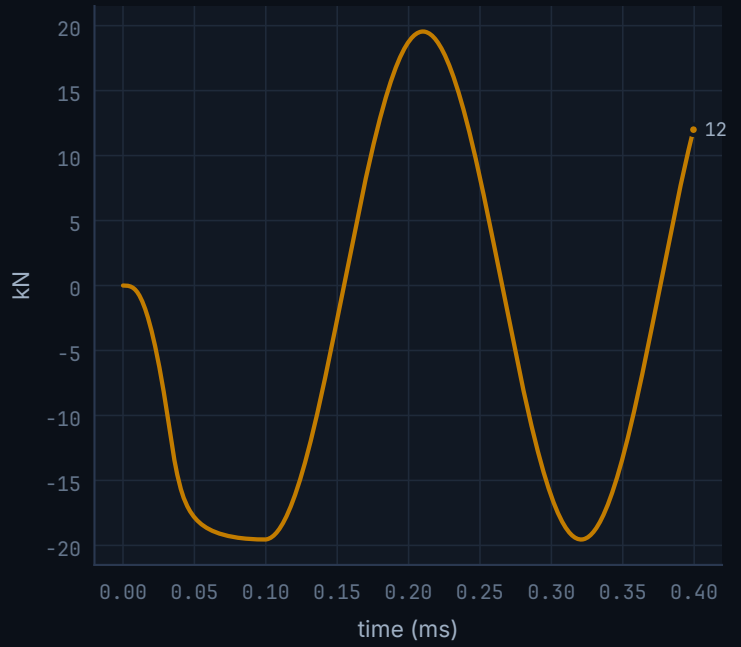
TRIAL 2: WEAPON SHAFT BEND

Time histories of the governing states (nominal case)

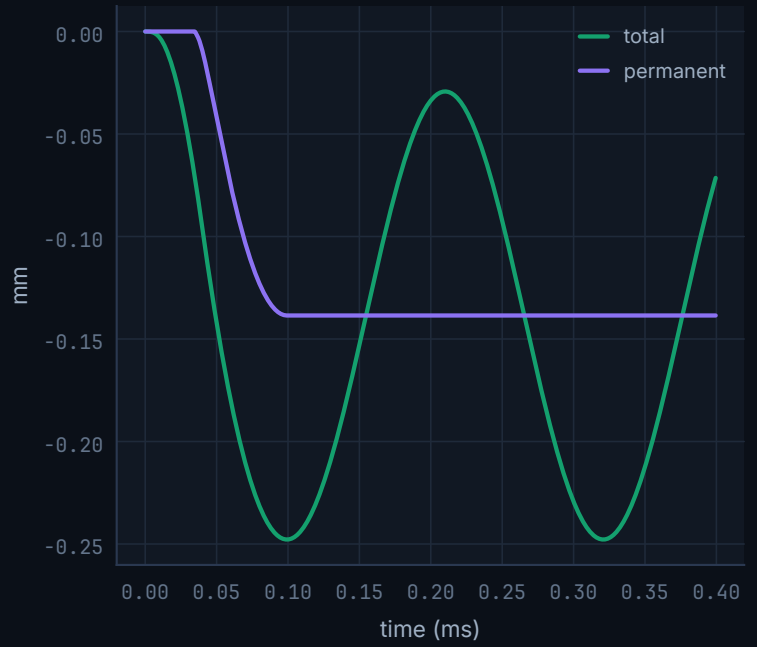
Tooth contact force



Shaft force



Shaft deflection and permanent set

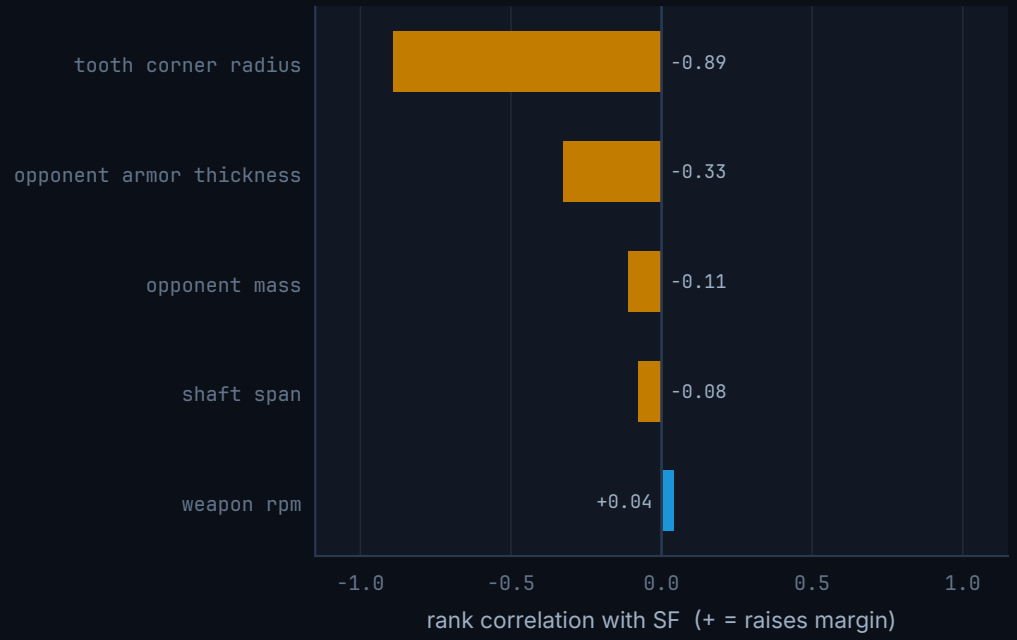


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TRIAL 2: WEAPON SHAFT BEND

What to design around: drivers of the governing margin and the design-variable sweep

Drivers of 'shaft bend' margin



Permanent bend vs shaft diameter and material



SIMBLOCKS // VERIFICATION

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TRIAL 2: WEAPON SHAFT BEND

● PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	3 bounds respected
stability_margin	● PASS	critical dt 2.78e-05 s / dt 2e-08 s = 1392.4x
dt_convergence	● PASS	shaft_set_mm: err 0.066% (order 1); F_contact_peak_kN: err 0% (order inf); F_shaft_peak_kN: err 0.0042% (order 1); E_to_opponent_J: err 0.057% (ord...
energy_ledger	● PASS	worst closure residual 0.015% across 3 dt levels (tol 5%)
limit:weapon not spinning → nothing happens	● PASS	matches expected limit
knockout:no shaft plasticity	● PASS	shaft_set_mm -100%, F_contact_peak_kN +0%, F_shaft_peak_kN +83%, E_to_opponent_J +0%
knockout:elastic contact	● PASS	shaft_set_mm +9.25e+03%, F_contact_peak_kN +1.22e+03%, F_shaft_peak_kN +22.4%, E_to_opponent_J +8.29e+03%
sensitivity	● PASS	5 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	150 samples, 0 failed finite/bounds/ledger (0.0%)
failure_modes	● WARN	shaft bend: nominal SF 0.722 FAILED, P(fail) 66.7%